

A decade ago, when his gated villa community in Bengaluru's Whitefield locality was grappling with a clogged drain, Tharun Kumar, 54, an IT security specialist, began looking for ways to set up a zero-maintenance sewage treatment method that could provide good-quality recycled water. The main concern, he says, was to prevent wastewater from draining into the nearby, much-polluted Varthur lake.

Inspiration came from nature itself—specifically, the stomach of the cow, which breaks down digestive matter through powerful anaerobic bacteria. The stomachs of ruminants comprise four chambers. Replicating this, Kumar and his team designed a septic tank with four stages.

"What's happening here is anaerobic digestion, that's a keyword," explains Kumar, who started ECOSTP in 2017. Anaerobic bacteria, found in a cow's gut, don't require oxygen. On the contrary, the aerobic bacteria in conventional sewage treatment plants (STPs) need air, which is provided through a blower running on a motor. Conventional STPs work well, says Kumar. But these require electricity to run the motors, a regular infusion of aerobic bacteria, and a full-time operator, all of which has a high cost.

Armed with grant and research inputs from the US-based Biomimicry Institute, ECOSTP designed a zero-power, zero-chemical septic tank, which can yield recycled water that meets the pollution control board's standards for flushing toilets and gardening. "A lot of science goes into this, though it is a simple thing," says Kumar, whose start-up currently has 235 clients across India, including

